Chlorophyll fluorescence and PAM fluorometry in microalgae

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<u>Precaution</u>:

- Stir regularly the sample with an inoculation loop
- Fill three 1ml cuvette with algae. Keep one for tests (sample 0), and two in the dark using aluminium foil (samples 1 and 2). Ensure the sample is free of air bubbles.

<u>Step 1</u>: **Learn how the machine works by splitting the job.**

Read the documentation briefly: 1 subgroup (1 or 2 persons) reads parts 4-7 and 9 (pages 13-24, 44-45); 1 subgroup (1 or 2 persons) reads part 11 (pages 48-60, 63-66). Then they share what they understood to perform the experiment they have been assigned to.

<u>Step 2</u>: **Test setting-up the PAM controller**

Using the PAM controller, do AUTO-ZERO (no sample) and check if fluo (with sample 0) is in the 200-500 range. If not, change PM-GAIN.

A typical 1s SAT plot would look like the one below (Figure 1)

Goal was then to check what time it takes to reach the plateau. To do that, the group did consecutive saturating pulses (~1 min apart), with different time for saturation.

SAT WIDTH [s]	F0	Fm	PS Yield
2	205	720	0,715
1,6	192	651	0,705
1	189	610	0,690
0,5	187	618	0,697
0,2	190	522	0,636

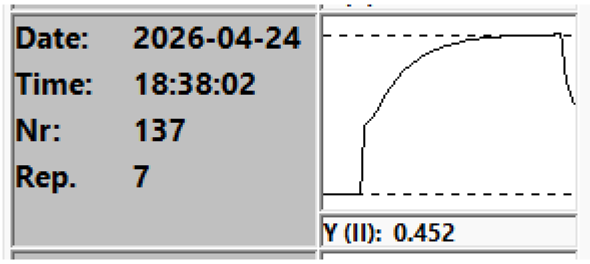
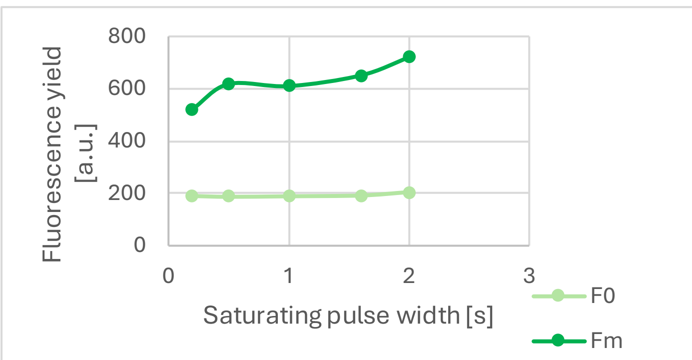
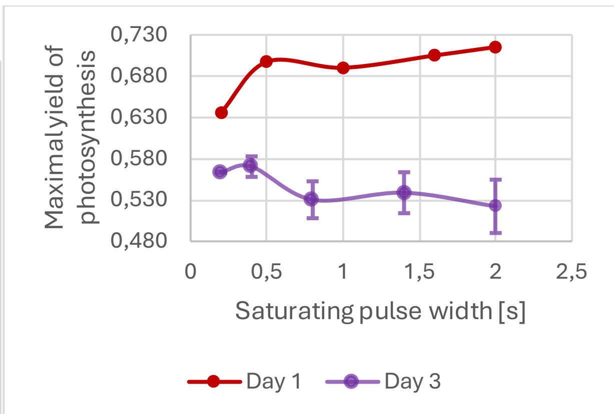


Figure 1: typical fluorescence yield upon saturating pulse stimulation





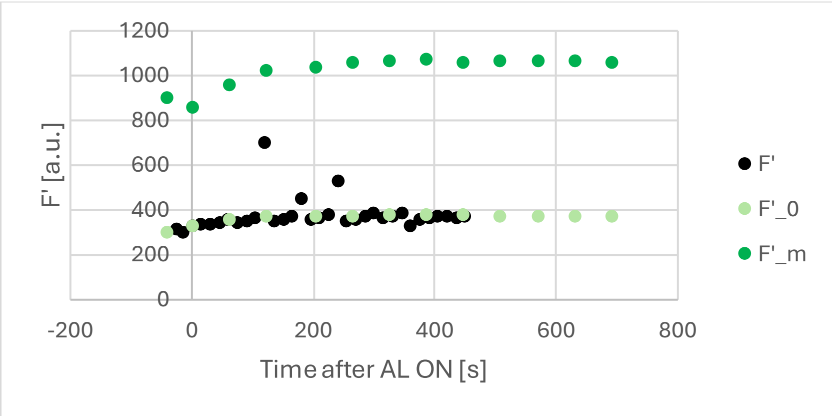
It appears that > 0.5 s seems enough to reach a plateau around $\Phi_{PSII} \simeq 0.7$ (red, right figure). While it worked really well on day 1, the group on day 3 (purple, right figure) has used a too tired cell culture. They repeated several times their measurement to see that their measurements were all giving results in the same range.

<u>Conclusion 1: For next experiments, the saturating pulse will be set to 1s.</u>

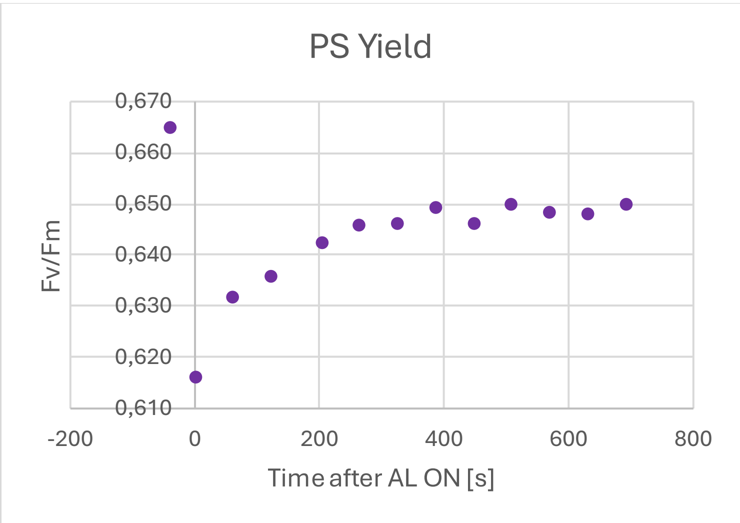
<u>Step 3</u>: **Perform an induction curve from dark-adapted cells (sample 1)**

Pick one AL of your choice. Once ready, setup sample 1 in the PAM, making sure to limit its exposure to light. AL should be off at this stage. In case, you could leave the sample 10 minutes in the dark before starting the experiment.

Run an induction curve (during ~10 minutes). Take F'-values on-the-fly. Remember to stir from time to time (not during SAT!). At the end, collect F_0, F_M and F'_M stored on the controller. Plot $F'(t)$, F'_0 and $F'_m(t)$.



It takes ~5 minutes for F'_m to converge. A closer look to the operating $\Phi_{PSII}(t)$ for this given light intensity is given below:



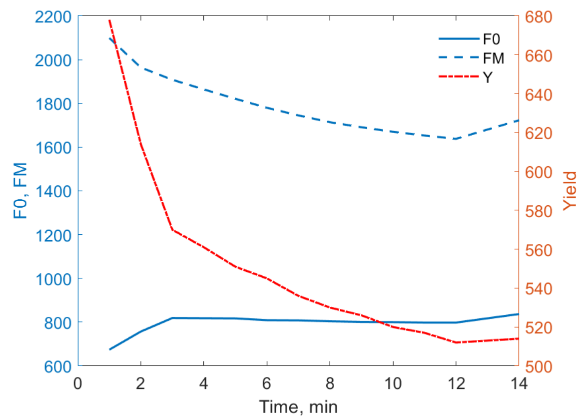


Figure 2: Fluorescence yields (left labels) and yield (per mil, right label) from group on Day 3. The first time point corresponds to no AL.

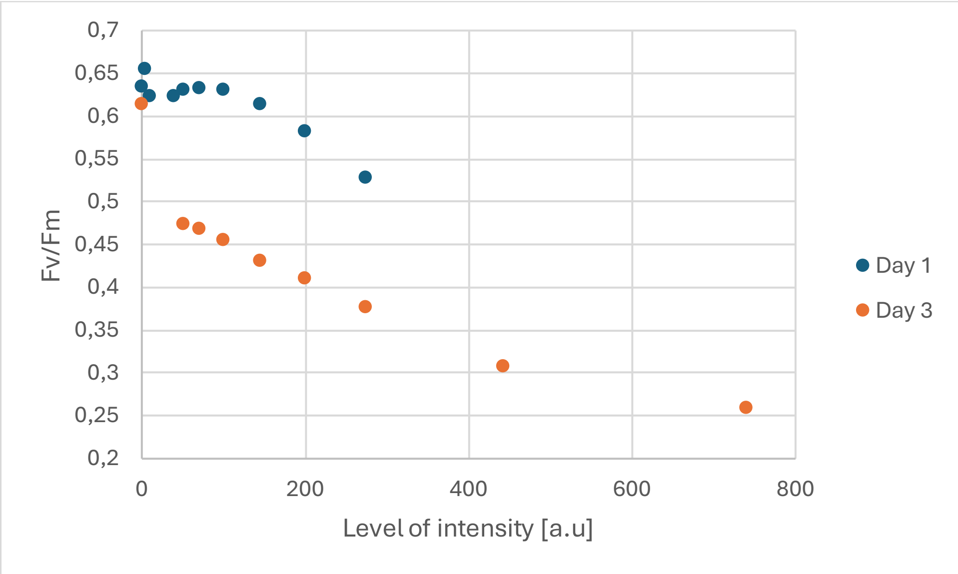
Similarly for the Day 3 experiments (with a new fresh culture),
while the relaxation somehow is opposite, the relaxation time is similar.

In all cases, before the AL is ON, Φ_{PSII} is the highest. It takes > 5 minutes to reach the plateau.

<u>**Conclusion 2: For the next experiment, the cell will be light adapted for 6 minutes.**</u>

Step 4: **Perform a light curve from dark-adapted cells (sample 2).**

Similar procedure than step 3. The goal here is to get $\Phi_{PSII}^{max}(I)$, for different actinic light intensities. It was done using the LIGHT CURVE mode. At the end, collect F_0, F_M and F_M' stored on the controller. The next plot displays the maximal photosynthetic rates $\Phi_{PSII}^{max}(I)$.

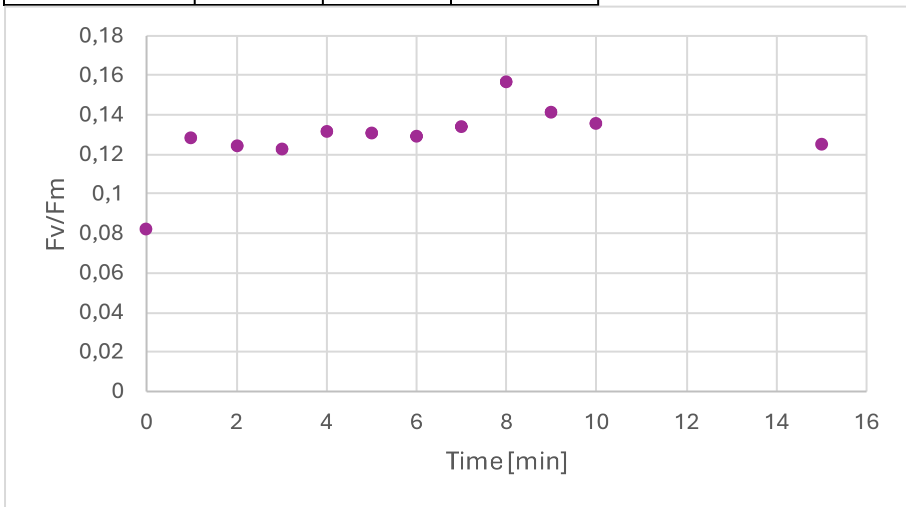


Conclusion 3: This shows that, for both days, maximal yield of photosynthesis decays by increasing the light intensity.

Step 5: **Perform a recovery curve from cells exposed to strong sunlight (all afternoon).**

Learning from the results above, the group decides to perform saturating pulses every minute for 10 minutes.

Timing (minute)	F0	Fm	Fv/Fm
0	434	473	0,082452
1	400	459	0,12854
2	392	448	0,125
3	385	439	0,123007
4	374	431	0,132251
5	371	427	0,131148
6	370	425	0,129412
7	367	424	0,134434
8	348	413	0,157385
9	363	423	0,141844
10	369	427	0,135831
15	361	413	0,125908



This graph shows that the yield of photosynthesis for this strongly exposed sample is way lower (around 0.1) than the dark-adapted ones (0.5 to 0.7). It may increase over time, but on a time scale not really accessible on the timing of the workshop. Actually, depending if cells have been photodamaged, it can take hours to repair.